Bom of KidBright Net 2 V1.0 20250501							
Item	Designator	Value	Package	'Description	Part Number	Manufacturer	Qty
1	BT1	Holder2032	Hold Vertical	Battery Holder Vertical Type for CR2025,CR2032	BS-5	GLD	1
2	BZ1	KW12M85A05	PCB Hold	Magnetic Buzzer,5Vp-p/50mA(max),Dia. 12mm, Magnetic Buzzer,5Vp-p/50mA(max),Dia. 12mm, Square Wave/10cm,Coil Resistance 45Ω±15%	KW12M85A05	KINGWEI	1
3	C1, C2, C3	1nF	0603[1608]	CHIP CAP 1nF/100V, +/-10%	06031C102KAT2A	AVX Corporation	3
4	C4, C5, CIN	1uF/10V	0603[1608]	Chip Cap 1µF/10Vdc, ±10%, X7R	0603B105K100CT	WALSIN	3
5	C6, CL	47uF/10V	0805[2012]	Chip Monolithic Ceramic Cap 47µF/10Vdc, ±20%, X5R	GRM21BR61A476ME15L	MURATA	2
6	C7, C9, C15, C16, C17	0.1uF	0603[1608]	Multilayer Ceramic Chip Cap 0.1µF/50Vdc ±10%, X7R	CL10B104KB8NFNC	SAMSUNG	5
7	C8, C18	10uF/25V	0805[2012]	Multilayer Ceramic Chip Cap 10µF/25Vdc, ±10%X5R	CL21A106KAYNNNE	SAMSUNG	2
8	C10, C11	10pF	0603[1608]	Multilayer Ceramic Capacitor 10pF/50Vdc ±5% COG	CL10C100JB8NNNC	SAMSUNG	2
9	C12, C13	0.47uF/16V	0603[1608]	MURATA MURATA, MURATA	GRM188R71C474KA88D	MURATA	2
10	CON1	SHT31 (male)	PIT 3.5 mm(male)	Plug-in Terminal Block 4pins,Male,300V/8A Pitch 3.5mm	15EDGVC-3.5-04P-14-00AH	DEGSON	1
11	CON1-1	SHT31 (female)	PIT 3.5 mm (female)	Plug-In Terminal Block 4 Poles,Female,300V/8A Pitch 3.5mm	15EDGK-3.5-04P-14-00AH	DINKLE	1
12	CON1-1	SHT30	2.00MM	WAFER CONNECTOR 2.00MM 4 PINS,	HY2.0-4P	HY2.0-4P	1
13	CON3		2.54 mm Pitch	WAFER Connector 4 Pins,Pitch 2.54mm, Straight Type Single Row	600104ST	GTK	1
14	CON3-1	8 pin 1.25mm.	1.25mm Pitch 8 pin	JST Plug Connector ซ็อคเก็ต Socket หมุดต่อสาย Crimp Terminal Micro JST1.25 Micro JST1.25		GTK	1
15	D1, D2	1N4148W-7-F	SOD-123	Fast Switching Diode 100V/300mA(Forward Continuous Fast Switching Diode 100V/300mA(Forward Continuous	1N4148W-7-F	DIODES INC.	2
16	J3	Header 4 Female	2.54mm Pitch	P.C.Board Socket 1x4Pins Female Header Single Row, Pitch 2.54mm	FHSS04G10/RH	GTK	1
17	J4	Pinhead 5Pins Duble [5x2]	2.54 mm. Pitch	Pin Header Connector 5 x 2 , Straight Type, Single Row, Pitch 2.54mm, Gold Plated	2211S-05G	NELTRON	1
18	J7	Wind	Through Hole (RJ11-6P4C)	Modular Jack Female 6P4C,RJ11,Top Entry Vertical	MJK607-P641G15F	MMT	1
19	J8	Rain	Through Hole (RJ11-6P4C)	Modular Jack Female 6P4C,RJ11,Top Entry Vertical	MJK607-P641G15F	MMT	1
20	LD1	Blue	1206 (3216)	CHIP LED,Blue Color 160-350mcd Water Clear Lens	HL-PC-3216H203BC	HONGLITRON IC	1
21	LD2	RED	1206 (3216)	HL-PC-3216S9AC Water Clear Lens,Viewing Angle 140 Degree	HL-PC-3216S9AC	HONGLITRON IC	1
22	LD3	Amber	1206 (3216)	OASISTEK Water Clear Lens, Luminous Intensity 66 to 190mcd, Water Clear Lens, Luminous Intensity 66 to 190mcd,	TO-3216BC-MAE-NV	OASISTEK	1
23	LD4	Green	1206 (3216)	HONGLITRONIC HONGLITRONIC	HL-PC-3216U51GC	HONGLITRON IC	1
24	LD5	White	1206 (3216)	HONGLITRONIC HONGLITRONIC Yellow Diffused Lens,Viewing Angle 140 Degree Yellow Diffused Lens,Viewing Angle 140 Degree	HL-PC-3216H233W	HONGLITRON IC	1
25	LDR1	KE-10715	RADIAL 100 mil 2 Pin	PHOTO CONDUCTIVE CELL,GAMMA VALUE(0.7),10-15K OHM,-30+60CELCIS	KE-10715	WAITRONY	1
26	Q3	BC847B,215	SOT-23	Transistor NPN 45V/0.1A, Low Current Low Voltage General Purpose Switching and Amplification	BC847B,215	NXP	1
27	Q4	2N7002LT1G	SOT-23-3	Small Signal MOSFET N-Channel 60V/115mA,225mW RDS(on) 7.5Ω Max	2N7002LT1G	ON SEMICONDUCTOR	1
28	R1, R2, R3	330R	0603[1608]	RES SMD 330 OHM 5% 1/10W 0603	0603WAJ0331T5E	ROYALOHM	3
29	R4, R23	1.2K	0603[1608]	Thick Film Chip Resistors 1/10W,1.2KΩ ±5%	0603WAF1201T5E	ROYALOHM	2
30	R5, R6, R7	220R	0603[1608]	RES SMD 220 OHM 5% 1/10W 0603	0603WAJ0221T5E	ROYALOHM	3

31	R8, R11, R12, R13, R14, R16, R17, R19, R20, R21, R22, R25, R28, R33	10K	0603[1608]	Thick Film Chip Resistors 10KΩ ±1%, 1/10W	0603WAF1002T5E	ROYALOHM	14
32	R9, R32	10R	0603[1608]	RES SMD 10 OHM 5% 1/10W 0603	0603WAJ0100T5E	ROYALOHM	2
33	R10	1K	0603[1608]	RES SMD 1K OHM 1% 1/10W 0603	0603WAJ0102T5E	ROYALOHM	1
34	R24, RLIM	47K1%	0603[1608]	Thick Film Chip Resistors 47KΩ ±1%,1/10W	0603WAF4702T5E	ROYALOHM	2
35	R26	20K	0603[1608]	Thick Film Chip Resistors 1/10W,20KΩ ±5%	MCR03ERTJ203	ROHM	1
36	R27	3.9K	0603[1608]	Thick Film Resistors ResHighPower 0603 3K9 1% 1/4W TC100	0603WAF3901T5E	ROYALOHM	1
37	R29	100K	0603[1608]	RES SMD 100K OHM 1% 1/10W 0603	0603WAF1003T5E	ROYALOHM	1
38	R30, R31	0R	1608[0603]	RES SMD 0 OHM JUMPER 1/10W 0603	0603WAJ0000T5E	ROYALOHM	2
39	SW1, SW2	Tact Switch 12x12	SPST-2	WEALTH METAL WEALTH METAL	TC-12XA-X-A01-ROHS	WEALTH METAL	2
40	SW3	Tact Switch 6x6	SPST 6x6	Tactile Switches Tactile Switches	TC-0102-X -ROHS	WEALTH METAL	1
41	U1	connector for Module ESP32 38Pin	2.54mm Pitch 19 pin x 2	PIN HEADER CONNECTOR 19 x 2 PINS,PITCH 2.54MM, SINGLE ROW STRAIGHT TYPE	PHSS20G10	GTK	1
42	U2	LORA	2.54mm Pitch	P.C.Board Socket 1x4Pins Female Header Single Row, Pitch 2.54mm	FHSS04G10/RH	GTK	1
43	U2-1	LORA	2.54mm Pitch	P.C.Board Socket 1x3Pins Female Header Single Row, Pitch 2.54mm	2212S-03G-85	GTK	1
44	U3	BD2242G-GTR	SOIC-8	1ch Adjustable Current Limit High Side Switch ICs, 1ch Adjustable Current Limit High Side Switch ICs, 1ch Adjustable Current Limit High Side Switch ICs,	BD2242G-GTR	ROHM	1
45	U4	BU33SD5WG-TR	SOT23-5	CMOS LDO Regulators 1Ch 3.3V/500mA Accuracy ±2%, for Portable Equipment	BU33SD5WG-TR	ROHM	1
46	U5	MCP7940NT-I/SN	SOIC-8	Battery-Backed I2C Real-Time Clock/Calendar(RTCC) with SRAM 64Bytes	MCP7940NT-I/SN	MICROCHIP	1
47	U7	Buffer	SC70-6	TINYLOGIC UHS DUAL BUFFER,WITH SCHMITT TRIGGER INPUT	NC7WZ17P6X	FAIRCHILD	1
48	Y1	32.768KHz	SMD	Crystal Unit, FC-135R, Frequency 32.768000kHz Crystal Unit, FC-135R, Frequency 32.768000kHz	X1A000141000600	EPSON	1
49	PCB KidBright Net 2 V1.0 20250501			PCB 100 mm. x 100 mm. FR4 1.6 mm Soder mark : red Overlay : white			1
50	U1-1	Module ESP32 38Pin	ESP-32 แบบ 38 pin TYPEC	ESP32 WROOM-32 Development Board TYPE-C CH340C/ CP2102 WiFi+Bluetooth Ultra-Low Power Consumption Dual Core Wireless Module ESP-32 แบบ38พิน-type C	ESP-32แบบ38พิน TYPEC	ESP-32แบบ38 พิน TYPEC	1
51	usb type c data cable 2m. **Hoco X14**	cable usb c data 2 m.		สายชาร์จ 2m **Hoco X14** Speed Charger ยาว 2 เมตร สำหรับ Type C สีดำ	usb c cable **Hoco X14**	usb cable data	1
52	OlED 1.3" 4 Pin	OlED 1.3" 4 Pin	OlED 1.3" 4 Pin	โมดูลหน้าจอ 1.3 นิ้ว Oled ฟ้า 128X64Spi / Iic I2C	OlED 1.3" 4 Pin	OlED 1.3" 4 Pin	1
53	SHT30	SHT30	HY2.0-4P	โมดูลเซนเซอร์วัดอุณหภูมิและความชื้น Sht30 Iic I2C อินเตอร์เฟส Hy 2.0-4P	HY2.0-4P	SHT30	1
54	Battery CR 2032	CR2032 3V/225mAh	CR 2032	Lithium Coin Battery 3V/225mAh,Diameter 20mm	CR-2032L/BE	PANASONIC	1
55	Spacer M2.5 x11 mm.		Female to Female M2.5 x 11	M2.5 x 11 mm. Hex Brass Motherboard Standoffs Female Spacing Screws Metric Thread Board Rack Stud Column Bolt Spacers 1 order บรรจุ 30 ตัว			7
56	Screw M2.5 x 5 mm.		M2.5 x 5	Screw M2.5 x 5 mm. 1 order บรรจุ 500 ตัว			1
57	สกรูขันพลาสติก M3 x 6 mm.		M3 x 6	สกรู หัวกลม สกรู น็อต หัวกลม แฉก + ขันพลาสติก M3 x 6 mm. 1 order บรรจุ 400 ตัว			1
58	LoRa Module E28-2G4T12S			LoRa Module E28-2G4T12S			1